

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING

Course Code: FCC3242

LECTURE THREE

Sectional Views

Lecture Outline

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2.1 Introduction

Welcome to the third lecture which is the development of section views. In this lecture, we are going to discuss methods of drawing views of sections which are formed after dissecting solid objects in various directions. The purpose of cutting through the solids is to reveal the material of construction e.g. concrete, reinforcement steel, glass, water, sand, etc. found in many civil engineering objects. However, in this lecture we will only discuss the sectional views of basic objects and the techniques used to detail the drawings.

3.2 Learning Outcomes

At the end of this lecture, you should be able to:

- a) Draw the sectional views resulting from cutting through an object.
- b) Meet the quality expectations for the drawings produced

3.3 Introduction to Sectional Views

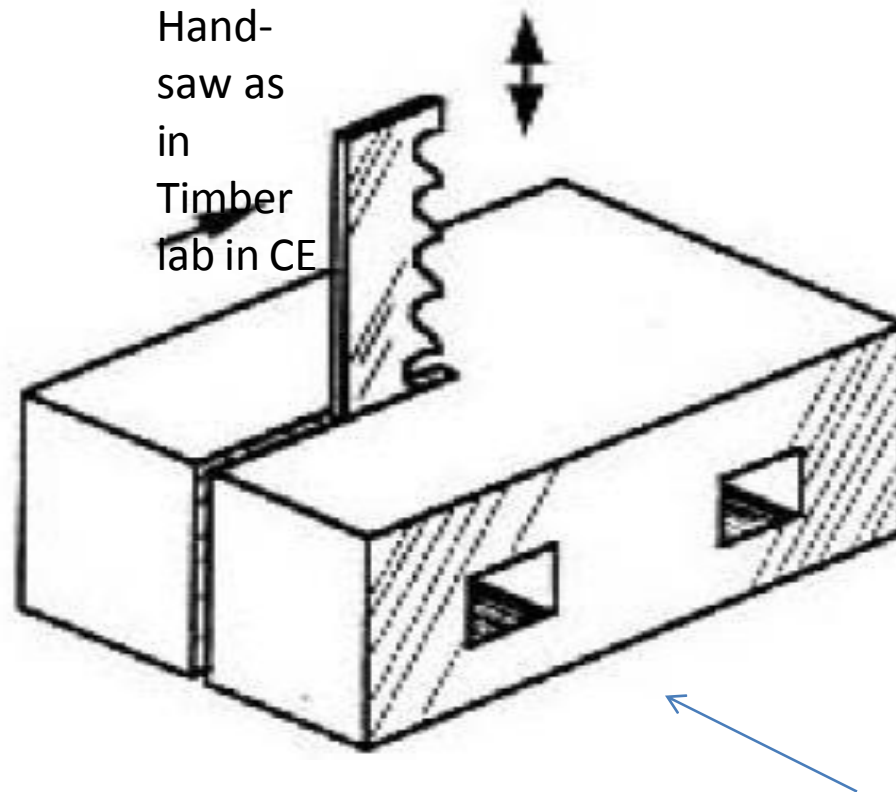
An orthographic projection of an object will show the exterior surfaces. If there are interior details and are hidden they are represented with the hidden-line symbol (thick dashes).

Purposes of Section Views

- a) Clarification of the views by
 - i. Reducing or eliminating the number of the hidden lines especially when the hidden features are complex.
 - ii. Revealing the cross section's shape and material.
- b) Facilitating the dimensioning of the hidden detail.

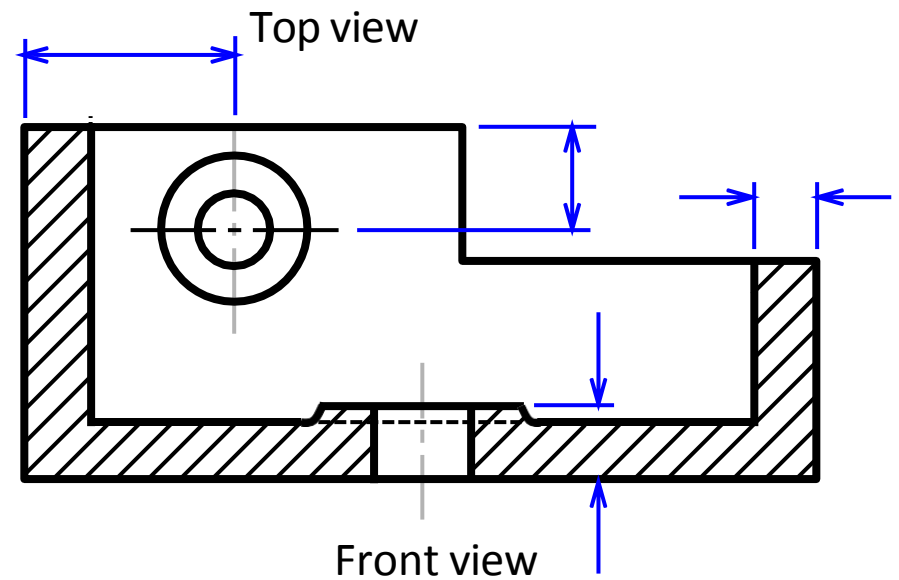
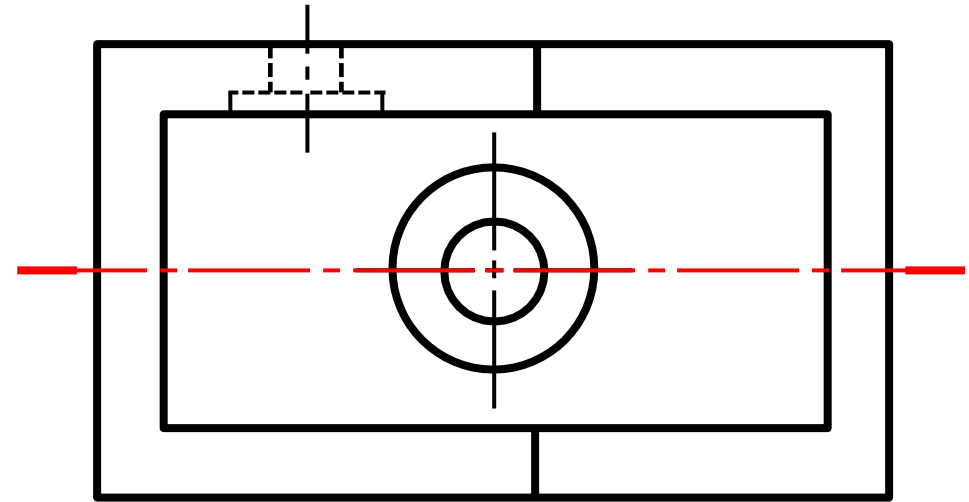
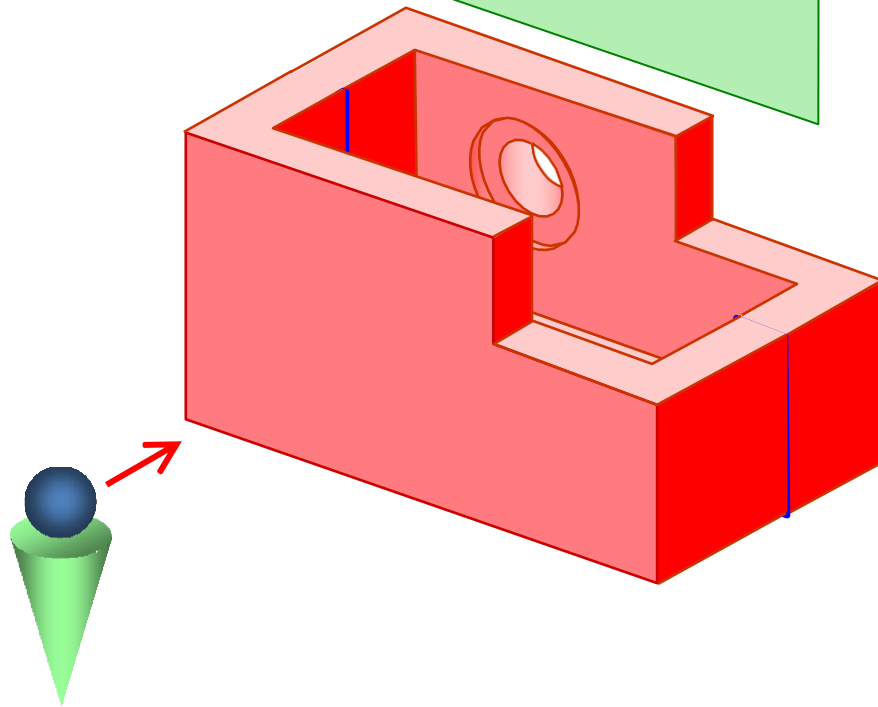
3.4 Procedure of getting section views

Imagine slicing the object in the middle as shown in the figure below



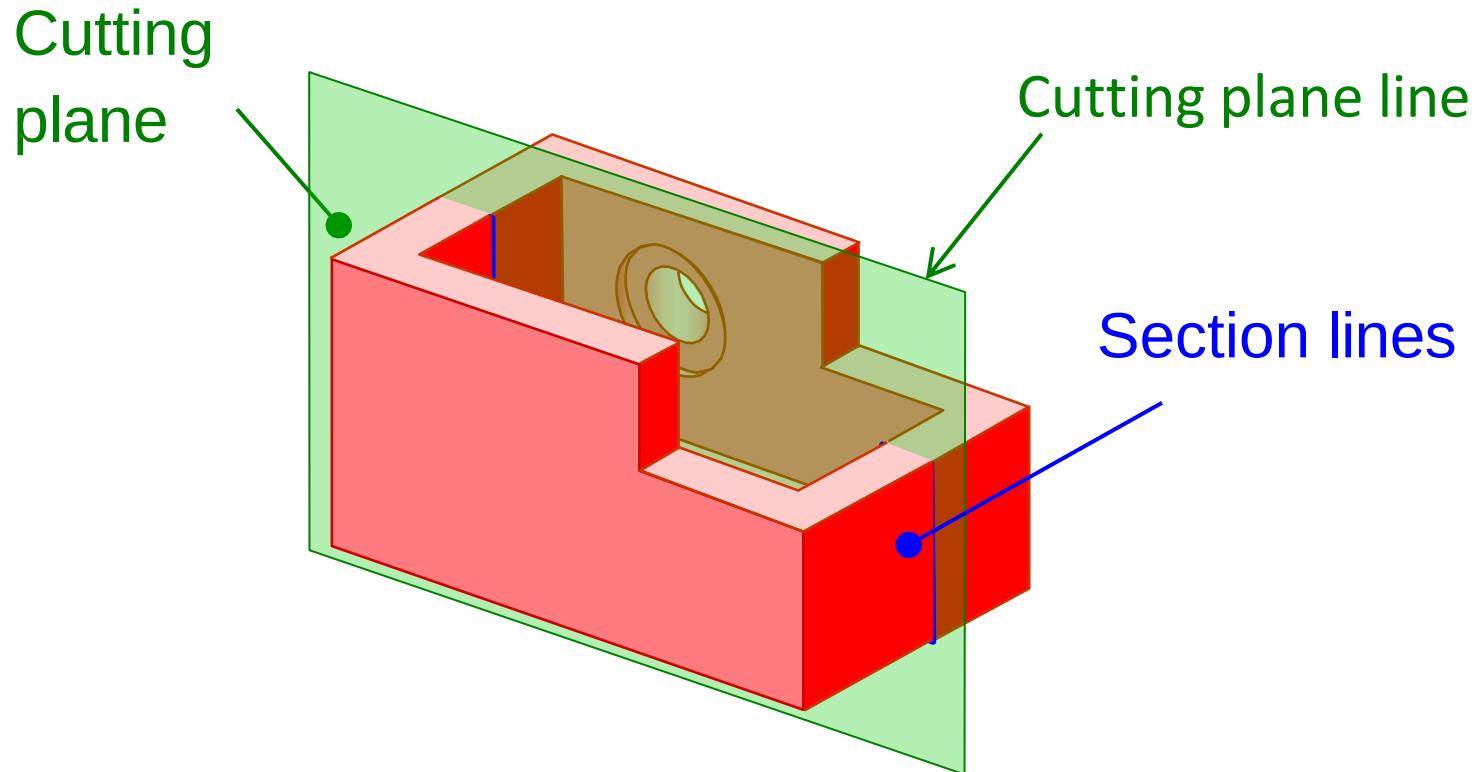
- a) A sectional view is made by passing an imaginary cutting plane through the object. One part is removed, exposing the interior details to view. The location of the imaginary cutting plane is shown on a drawing with a cutting-plane line.
- b) A sectional view frequently replaces one of the regular views. For example, a regular front view is replaced by a front section

Obtaining a section view (continued). NB: The views are third angle orthographic projection

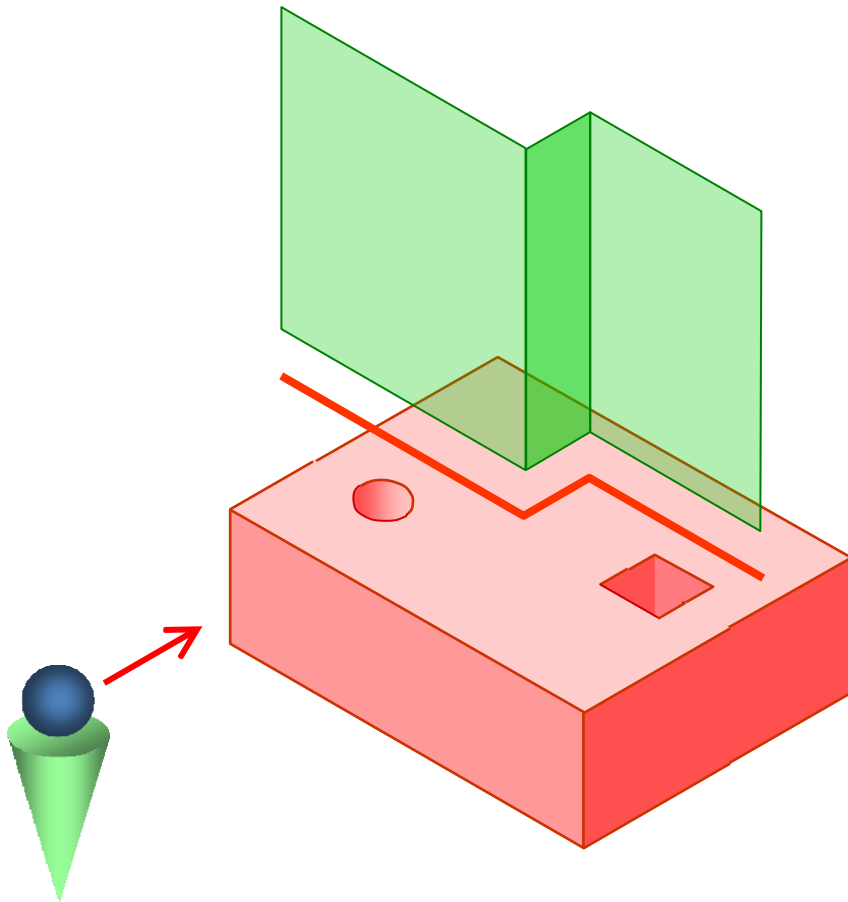


3.5 Terminologies used in sectional views

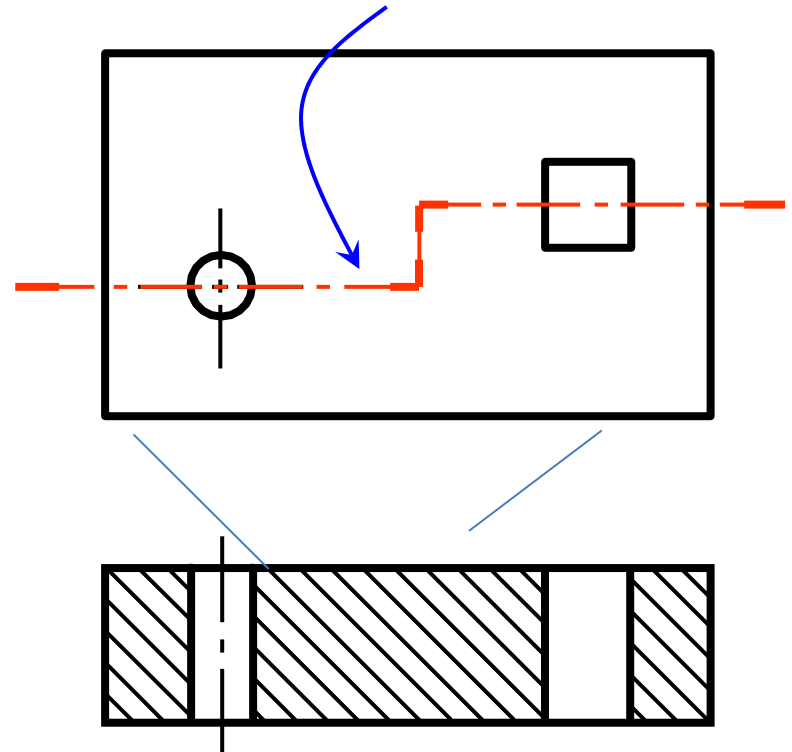
Cutting plane is a plane that ***imaginarily cuts*** the object to reveal the internal features.



Cutting plane line is an *edge view* of the cutting plane.

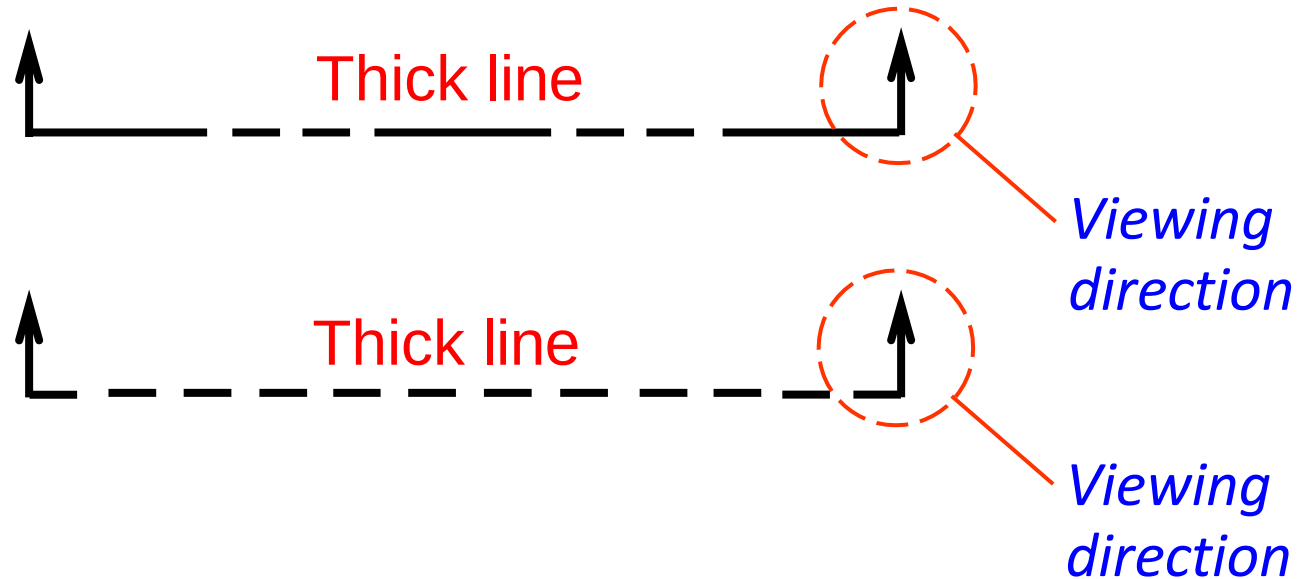


Indicate the *path*
of cutting plane.

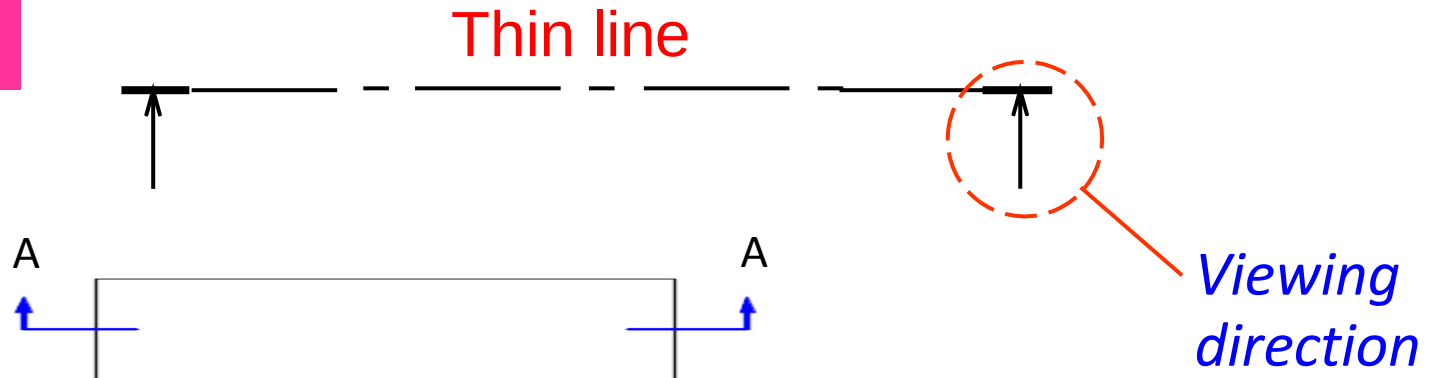


3.6 Cutting Plane Line styles

ANSI
standard



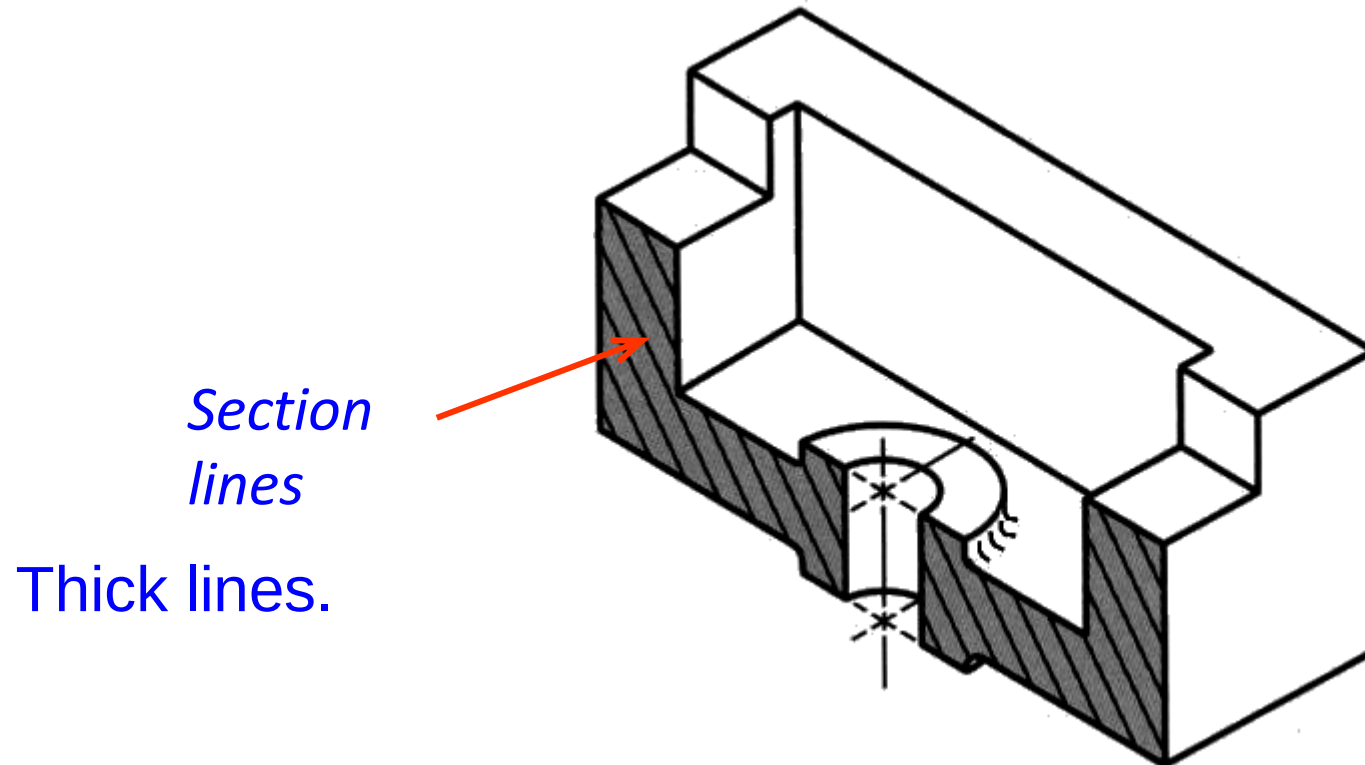
JIS & ISO
standard



This course

3.7 Section Lining

Section lines or **cross-hatch lines** are used to *indicate the surfaces that are cut by the cutting plane.*



General purpose symbol: the cast iron symbol below is used most often for any materials.



Cast iron,
Malleable iron

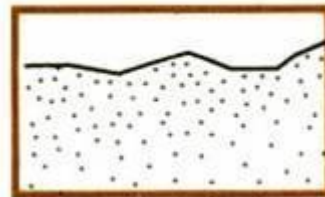
Other types of sectioning (hatching)



Steel



Concrete



Sand



Wood

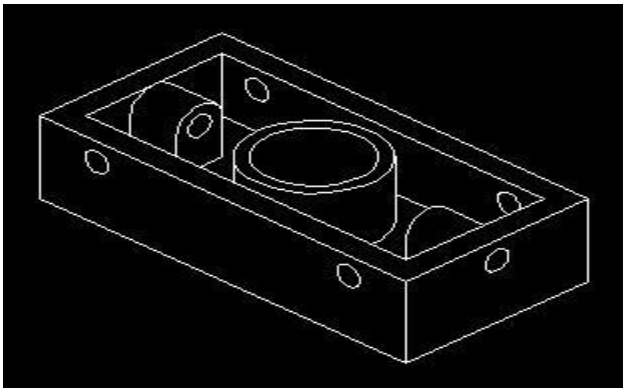
3.8 Kinds of Sections

1. Full section
2. Offset section
3. Half section
4. Broken-out section
5. Revolved section (aligned section)
6. Removed section (detailed section)

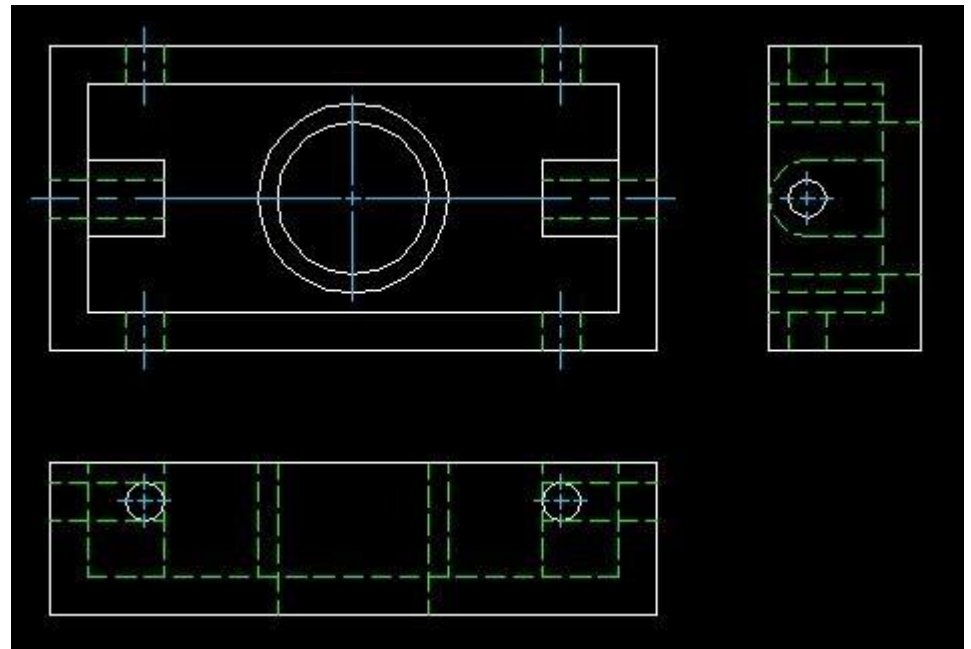
3.9 Drawings Containing Two or More Sections

Consider the object whose pictorial view is shown below:

Drawing the object using only orthographic views (to 3rd angle) yields the views shown on the right:

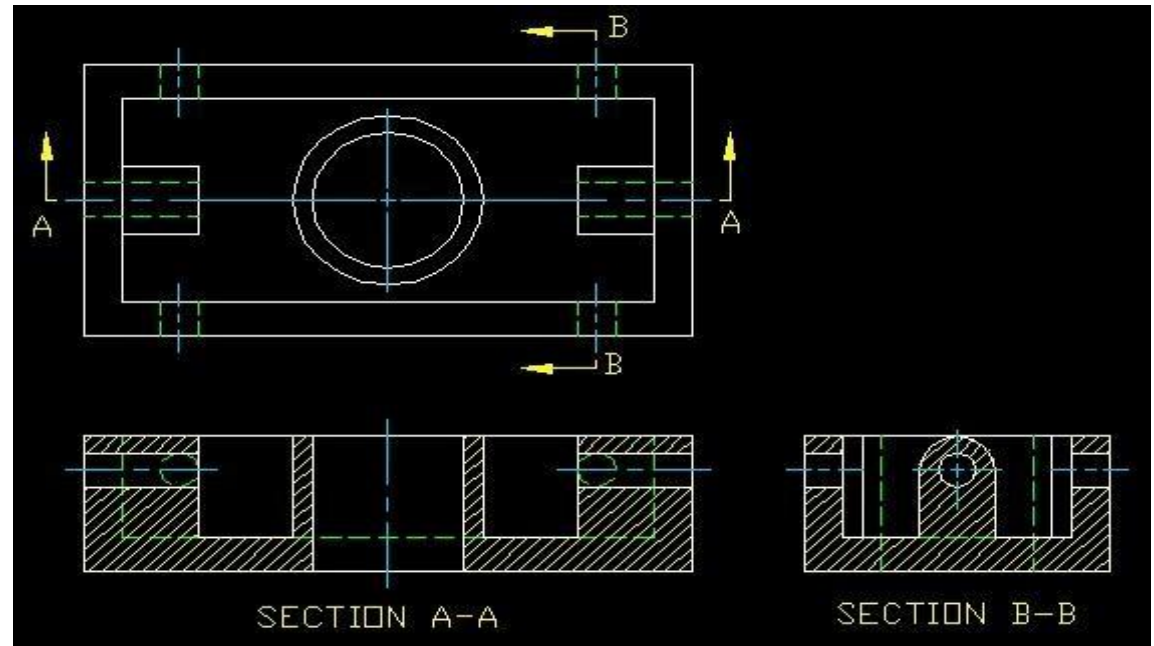
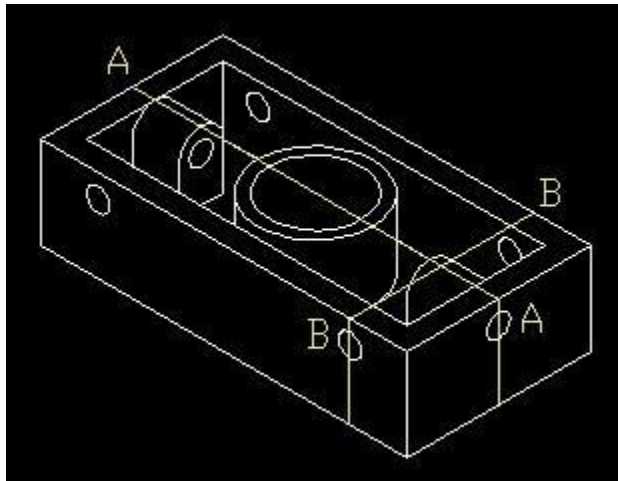


Note the number of hidden lines



Now cut the object in two strategic locations (cut lines **A-A** and **B-B**).

The full sectional views generated are more useful and definitive.

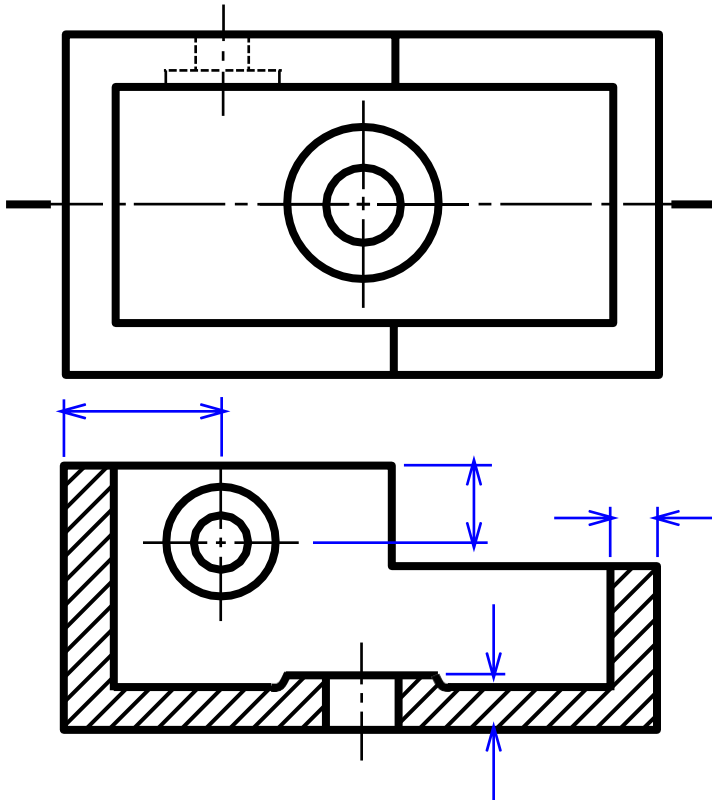


- a) Full sections in place of orthographic views makes the object easier to visualize
- b) Note that there are still some hidden lines remaining but very few
- c) Because this drawing contains two sections, section labels are now specified as **SECTION A-A** and **SECTION B-B**.

3.10 Dimensioning

In most cases, dimensioning of the section views follows the typical rules of dimensioning.

Good dimensioning practice on the section view



3.11 Summary

We have come to the end of lecture three. This lecture started by stating the aims of sectional drawing. We then discussed procedure of obtaining sections and how to present and detail these views.

3.12 Self - Assessment

3.13 References and further reading

References for further reading were listed in Lecture one. You are free to look at any of them for the purpose of this lecture.